Test report

This report lists the checks run on this clean-room reimplementation of Click to confirm it behaves like the original. All numbers were measured on Linux with Python 3.13.

Method

1. The package was written first, without reading Click's code. That version is the first commit (`d1a3a1e`).
2. Only then was Click's own test suite fetched (the `tests/` directory of the upstream `main` branch, which targets Click 8.6, plus its pytest configuration). The upstream source was never downloaded.
3. The suite was run, failures were analysed from the tests' expectations, the code was fixed, and the suite was re-run until it was clean.

Result of the compatibility suite

Outcome	Count
Passed	2262
Skipped (Windows-only tests)	24
Expected failure (marked `xfail` upstream)	1
Failed	0
Deselected (stress tests, off by default)	31000

The same result was obtained running from the source tree and from a clean virtual environment with the package installed. The commit tested is the one published on GitHub.

Progress during development

Stage	Passed	Failed
First full run (after adding the missing internals the tests import)	2031	230
After styling, pager, prompt and utility fixes	2186	76
After the test runner, version option and help fixes	2222	40
After exit handling, context resources and type fixes	2255	7
After the last fixes	2262	0

Results by test file

Test file	Passed	Skipped	Expected failure
`test_options.py`	699		
`test_termui.py`	259	23	
`test_arguments.py`	206		
`test_compat.py`	175		
`test_basic.py`	109		
`test_commands.py`	103		
`test_deprecations.py`	66		
`test_shell_completion.py`	58		
`test_defaults.py`	46		
`test_testing.py`	44	1	
`test_info_dict.py`	38		
`test_formatting.py`	37		
`test_context.py`	30		
`test_stream_lifecycle.py`	24		
`test_chain.py`	15		1
`test_command_decorators.py`	9		
`test_abort_interrupt.py`	8		
`test_parser.py`	7		
`test_custom_classes.py`	6		
`test_normalization.py`	3		
`test_imports.py`	1		
`test_exceptions/`	1		
`test_types/` (8 files)	68		
`test_utils/` (13 files)	250		
Total	**2262**	**24**	**1**

What the main files cover

- **Options and arguments** (`test_options` , `test_arguments` , `test_defaults`): declarations and naming rules, defaults and "unset" handling, flags and competing flags that share a name, environment variables, `multiple` and `nargs` , help text, deprecation labels.
- **Commands and context** (`test_commands` , `test_context` , `test_chain` , `test_basic`): groups, chained commands, context objects, resources that close with the context, parameter sources, the help option.

- **Terminal interaction** (`test_termui` , `test_utils`): prompts and confirmations, hidden input masking, styling, progress bars, the pager, the editor, `echo` , file helpers.
- **Help output and errors** (`test_formatting` , `test_info_dict` , `test_compat`): exact help layout, usage errors, ANSI-aware wrapping.
- **Shell completion** (`test_shell_completion`): Bash, Zsh, Fish and PowerShell scripts and results.
- **Test runner** (`test_testing` , `test_stream_lifecycle` , `test_abort_interrupt`): `CliRunner` , file descriptor capture, stream ownership, behaviour on repeated Ctrl-C.
- **Types** (`test_types/`): `Choice` , `File` , `Path` , ranges, conversion.
- **Compatibility** (`test_deprecations` , `test_imports`): deprecated names still work with warnings, and `import click` stays light.

Stress tests

These are off by default upstream. They were run explicitly:

Test group	Cases run	Passed	Failed
Pager cleanup (`test_echo_via_pager.py`)	1000 (all)	1000	0
Stream handling (`test_stream_lifecycle.py`)	1500 of 30000 (every 20th)	1500	0

The first sample found a real bug: the pager stress test failed in all 50 sampled cases because an internal module was not loaded on import. It was fixed and all 1000 pager cases then passed. The remaining 28,500 stream-handling cases were not run.

Other checks

- **Smoke test:** a small group with options, arguments and a subcommand was run through `CliRunner` for help pages, usage errors, unknown commands and invalid choices, and the output was inspected by hand.
- **Static check:** `pyflakes` over the package reports nothing except the intentional re-exports in `__init__.py` .
- **Packaging:** the package builds and installs into a fresh virtual environment, and the suite passes against the installed copy.

Limits of this evidence

- The suite is the strongest check available, but it only confirms behaviour the tests look at. Behaviour outside the tests could still differ.
- The 24 skipped tests only run on Windows. The Windows-specific code (key reading, editor paths) has not been executed.
- The target is upstream's development branch (Click 8.6), not the released 8.5.0. A few behaviours differ between those two.
- A side-by-side comparison with real Click 8.5.0 (same programs and inputs, diffing output and exit codes) has not been done.